Riding Wavelets: Reproducing “Riding Wavelets” on Open Data

Reproducible research (ML for Time Series)

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Why this project?

- We do not work in finance: curious on **very noisy** data and **non-trivial spectral analysis** with extreme events.
- We reproduced a hedge-fund preprint: **Bouchaud et al., "Riding Wavelets"** (jump classes via wavelets + Kernel PCA).
- Key motivation: **non-stationary markets** (volatility clustering, intraday patterns) so global templates (e.g., dictionary learning) fail.
- Goal: reproduce the full pipeline from the paper with **open data**, end-to-end, and test across **time scales** and **markets**.

Outline

Method > Data > Results

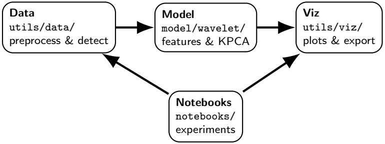
Reproducibility and code structure

Engineering (open + reproducible)

- Worked 50/30 hours, wrote 2k LOC, 30 notebooks, 73 commits, We open-source the code.
- **We re-implement the full pipeline.**

1-liner installation

```
chmod +x launch.sh && ./launch.sh
```



Code map (where things live)

- 5GB Stooq OHLC. 5-min, hourly, daily.
- Poland, Hungary, HK, US, UK Equities

Jump normalization (seasonality + local volatility)

- Markets highly non-stationary = standardize returns to jump score $x(t)$

$$x(t) = \frac{r(t)}{f(t)\sigma(t)}.$$

Seasonality f_t

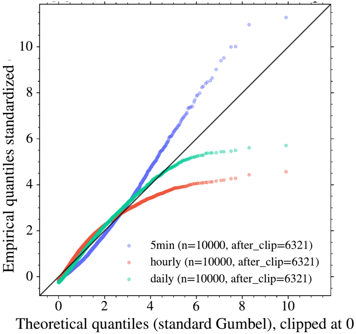
$$r_t = \log p_t - \log p_{t-\Delta}, \qquad \tau(t) = \text{time-of-day of } t,$$
$$m(\tau) = \mathrm{median}\Big\{|r_t| : \tau(t) = \tau\Big\},$$
$$q = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}} m(\tau)^2},$$
$$f_t = \frac{m(\tau(t))}{q}, \qquad r_t^{\mathrm{des}} = \frac{r_t}{f_t}, \qquad (\text{daily: } f_t \equiv 1).$$

Local volatility σ_t

$$\mu_t = \alpha r_t^{\mathrm{des}} + (1-\alpha)\mu_{t-1},$$
$$v_t = \alpha\left(r_t^{\mathrm{des}} - \mu_t\right)^2 + (1-\alpha)v_{t-1},$$
$$\sigma_t = \sqrt{v_t}, \qquad \alpha = \frac{2}{s+1}.$$

Gumbel diagnostic

Hypothesis: Jump scores are approximately max-stable (Gumbel-like) after normalization; we threshold with $\theta = 4$ (5-min).
Plot: Q–Q of $|x(t)|$ vs Gumbel.
What we see: Overall good fit; mild 5-min Fréchet deviation.



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Wavelet features and interpretable directions

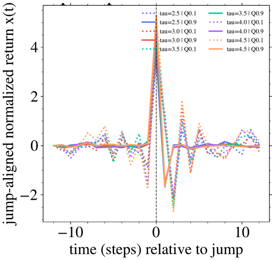
- We align jump signs ($t = 0$).
- We compute complex wavelet scattering.

$$\Phi(x) = \big(W_j \bar{x}(0),\ W_{j_2} \left|W_{j_1} x\right|(0)\big)_{1 \le j \le J,\ 1 \le j_1 < j_2 \le J}.$$

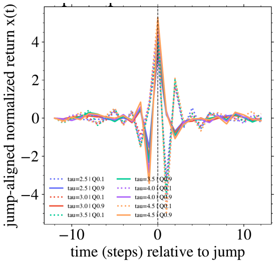
- We run Kernel PCA on feature vector $\Phi(x)$: $K(\Phi(x_i), \Phi(x_j))$.

Research question 1: robustness to the threshold (5-min)

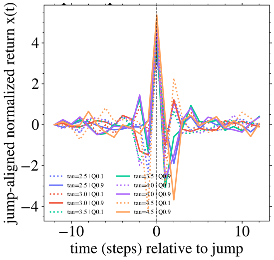
Hypothesis: Average jump profiles stay stable across θ.
Plot: Overlay mean profiles across θ (5-min).



D_1 overlays



D_2 overlays



D_3 overlays

What we see: D_1 and D_3 stable; D_2 noisier at extremes.

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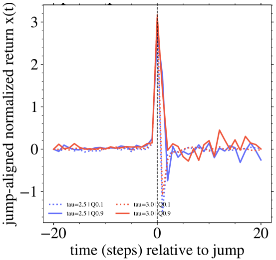
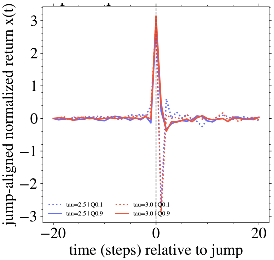
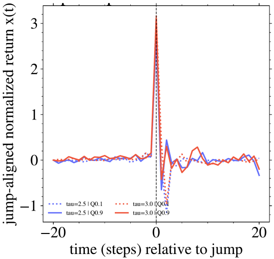
Hypothesis: Asymetry/Mean-Revert/Trend structure
Plot: Overlay mean profiles across θ (daily).



D_1 (daily)



D_2 (daily)



D_3 (daily)

What we see: Asymetry, less daily data = higher noise.

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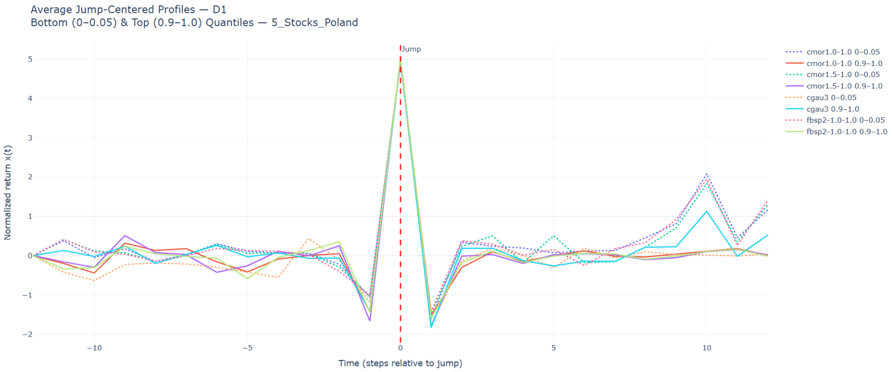
Research question 3: robustness (SS ablation + wavelets)

Hypothesis: Features are robust to (i) Scattering Spectra and (ii) wavelet choice.
Plot: Wavelet-family sweep for D_1; SS ablation runs (text).



What we see: With-SS vs no-SS: embeddings stay very similar; D_1 correlation stays high.

What we see: D_1 stays stable across wavelets (fbsp/cmor/cgau).

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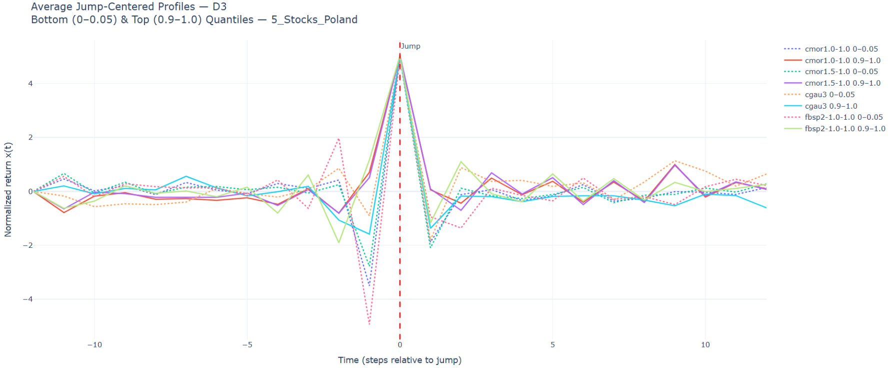
Trend direction: D_3

Hypothesis: D_3 is a stable trend direction.
Plot: D_3 profiles across wavelets / markets.



What we see: D_3 is unstable on PL/HU; we need more data (US placeholder).

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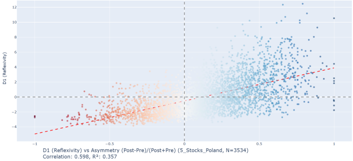
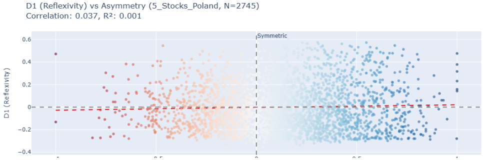
RQ4: Asymmetry – daily vs 5-min

Hypothesis: Daily asymmetry correlates; 5-min does not. **Plot:** Correlation (daily vs 5-min).



daily



5-min

What we see: Daily: strong asymmetry (macro), 5-min: weak.

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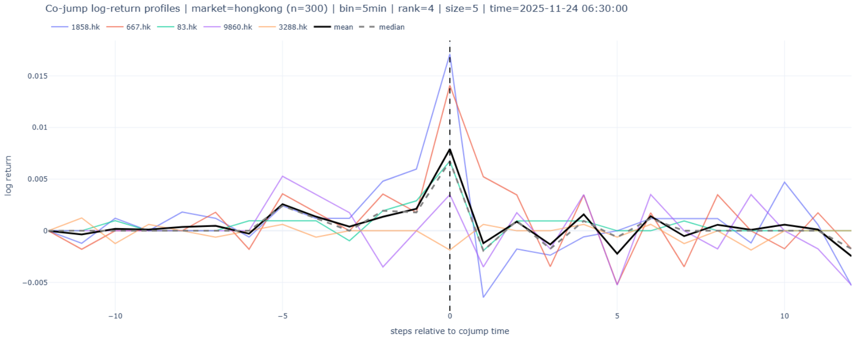
Research question 5: 5-min cojumps looks the same

Hypothesis: 5-min Jumps are endogeneous = Symetric.
Plot: Example of 5-min co-jumps.



What we see: Very symetric/reflexive jumps, no exogenous activity.

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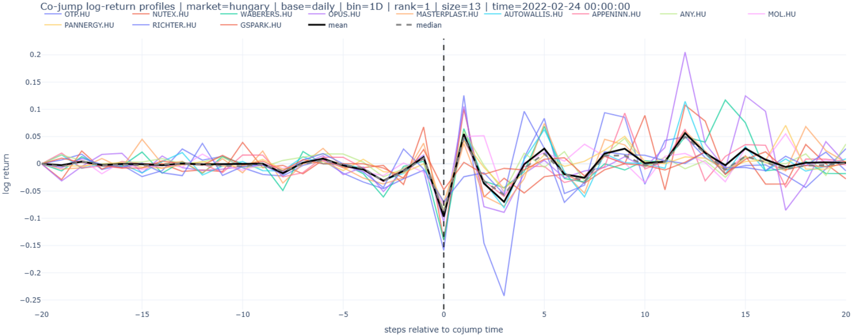
Research question 5: Daily co-jumps looks different

Hypothesis: Daily co-jumps show low reflexivity.
Plot: Co-jump indicator from aggregated D_1.



What we see: Reflexivity signal very low, exogeneous activity

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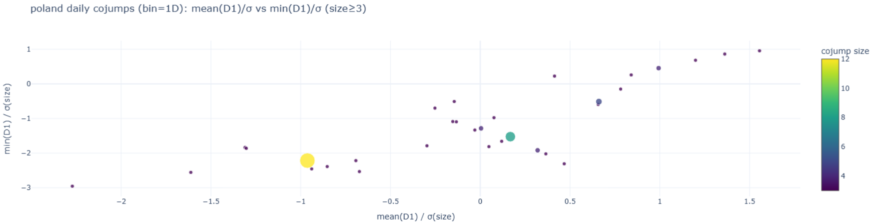
Co-jumps + takeaways

Hypothesis: Large daily co-jumps looks the same.
Plot: Global Summary - Cojumps similarity index.



What we see: They share same D1 values = looks the same

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Takeaway

Full pipeline, open data, unsupervised robust jump classes (heavy tails, stable profiles across thresholds/wavelets/markets), co-jumps consistents.